EduSaaS PDDD - Part 4

Chapters 18 to 25: Deployment, Operations, Security and Readiness

Document Type: Production Database Design Document (PDDD)

Schema: education | Database: PostgreSQL | Architecture: Backend-owned persistence, stateless AI services

Chapter 18: Complete Deployment Scripts

Deployment must follow dependency order: extensions, schema, enums, reference tables, master tables, dependent tables, learning, assessment, community, AI support, indexes, functions, triggers and seed data.

```
CREATE EXTENSION IF NOT EXISTS pgcrypto;
CREATE SCHEMA IF NOT EXISTS education;
-- Create enums, tables, FKs, indexes, functions, triggers, seed data in versioned scripts.
```


Chapter 19: Rollback Scripts

Rollback must execute in reverse order: triggers, functions, indexes, constraints, recommendation AI columns, AI tables and seed data. For major failures, prefer backup restore.

```
DROP TRIGGER IF EXISTS trg_recommendations_updated_at ON education.recommendations;
DROP FUNCTION IF EXISTS education.set_updated_at();
ALTER TABLE education.recommendations DROP COLUMN IF EXISTS predicted_rating;
DROP TABLE IF EXISTS education.course_ratings;
```

Chapter 20: Security Standards

Control	Requirement
Password storage	Hash only, Argon2id/BCrypt preferred
OTP storage	otp_hash only
Refresh token storage	token_hash only
Transport	TLS 1.2+
Authorization	RBAC, deny by default
Secrets	Externalized to Kubernetes/OpenShift/Vault
SQL injection	Prepared statements/ORM binding
Backups	Encrypted and access restricted

Chapter 21: Backup & Recovery

Area	Target
Full backup	Daily
Incremental backup	Every 4 hours
Transaction log backup	Every 15 minutes
RPO	15 minutes
RTO	1 hour
Retention	Daily 90 days, Monthly 1 year, Annual 7 years

Chapter 22: Performance & Scalability

Area	Target/Strategy
OLTP queries	< 100 ms
Dashboard queries	< 2 sec
Connection Pool	HikariCP 20-50
Caching	Redis for settings, courses, skills, tags
Partitioning	student_answers, activity_logs, login_history, notifications
High-growth tables	student_answers, activity_logs, login_history, community_comments

Chapter 23: ER Diagram Catalogue

The ER model covers 46 tables, 8 domains and 60+ relationships. Major parent entities are users, courses, skills, domain_roles and community_posts. Relationship patterns include one-to-many, many-to-many and self-referencing relationships.

Chapter 24: Data Flow Diagrams

Flow	Primary Tables
Authentication	users, roles, permissions, role_permissions, auth_otps, refresh_tokens
Learning	courses, lessons, enrollments, progress, certificates
Assessment	questions, quiz_sessions, student_answers, student_skill_results
Recommendations	activity_logs, login_history, course_ratings, recommendations
Community	community_posts, community_comments, reactions, tags, bookmarks
Employer	jobs, applications
Administration	notifications, subscriptions, reports, settings, announcements

Chapter 25: Production Readiness Checklist

Readiness Area	Required Status
Database	46 tables, enums, constraints, indexes, functions and triggers validated
Security	RBAC, password hashing, token hashing, TLS, secrets, audit logging enabled
Performance	Load/stress/endurance testing completed
Backup & Recovery	Restore, PITR and DR tested
Monitoring	Database, API, storage, errors and business KPIs monitored
Deployment	Dry run completed, rollback tested
Sign-off	DBA, Security, Architecture, QA, Ops and Business approved

Final Executive Conclusion

The EduSaaS PDDD provides a production-ready blueprint for database design, deployment, security, operations, recovery, performance, ER/DFD documentation and go-live governance. The architecture supports backend-owned persistence with stateless AI services and a PostgreSQL education schema containing 46 documented tables.